Worksheet 8-11 & 8-12 A Volumes: Washer Method #1
AP Calculus Name_____

Directions: Find the volume of the solid obtained by rotating the region bounded by the given curves about the specified line. Sketch the region, the solid, and a typical washer.

1.) $y = x^3,\ \ y = x,\ \ x \geq 0$ about the x-axis

2.) $y = 6 - x^2,\ y = 2$ about the x-axis

3.) $y^2 = x,\ x = 2y$ about the y-axis

4.) $x = 2 - y^2,\ x = y^4$ about the y-axis

5.) $y = x^2,\ x = y^2$ about $y = 1$

6.) $y = x^3,\ y = 1,\ x = 2$ about $y = -3$

7.) $y = x^3,\ y = 0,\ x = 1$ about $x = 2$

8.) $y = \sqrt{x+2},\ x = 2$ about $x = -3$

Answers:

1.) $\frac{4\pi}{21}$ 2.) $\frac{384\pi}{5}$ 3.) $\frac{64\pi}{15}$ 4.) $\frac{248\pi}{45}$ 5.) $\frac{11\pi}{30}$

6.) $\frac{471\pi}{14}$ 7.) $\frac{3\pi}{5}$ 8.) $\frac{544\pi}{15}$